



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Branch: CSE (2025-29)

Semester: 3

Subject: Programming with Java

ASSIGNMENT No.: 7

TITLE: : Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement:

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives:

To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory:

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

Outcomes:

Successfully implemented inheritance and interfaces in Java to achieve code reusability and abstraction. Applied these concepts to build real-world applications like a Shape calculator and an E-Commerce Product System.

```
interface Sound{
    void makeSound();
}

class Animal{
    String name;
    Animal (String n){
        this.name = n;
    }
    void eat(){
        System.out.println(name + " is eating.");
    }
}

class Dog extends Animal implements Sound{
    Dog(String n){
        super(n);
    }
    public void makeSound(){
        System.out.println(name + " says: Woof!");
    }
}

public class JAVA_EXP_7{

    public static void main(String[] args) {
        Dog dog = new Dog("Tommy");
        dog.eat();
        dog.makeSound();
    }
}
```

```
orkspaceStorage\6b4136d8ec875b8bb324a9029103fc22\redhat.
eabe1187\bin' 'JAVA_EXP_7'
Tommy is eating.
Tommy says: Woof!
PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>
```

Conclusion:

This assignment demonstrates how inheritance enables code reuse by allowing subclasses to acquire properties and methods from a parent class, while interfaces define contracts that ensure consistent behaviour across different classes. The Shape application and E-Commerce Product System successfully illustrate how these OOP concepts support abstraction, polymorphism, and modular software design in real-world scenarios.

Exercise:

1. Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

```
class Shape {
    void area() {
        System.out.println("Area method not defined for generic shape.");
    }
}

class Circle extends Shape {
    double radius;
    Circle(double radius) {
        this.radius = radius;
    }
    void area() {
        double result = Math.PI * radius * radius;
        System.out.println("Circle Area: " + result);
    }
}

class Rectangle extends Shape {
    double length, width;
    Rectangle(double length, double width) {
        this.length = length;
        this.width = width;
    }
    void area() {
        double result = length * width;
        System.out.println("Rectangle Area: " + result);
    }
}

public class JAVA_EXP_7 {
    public static void main(String[] args) {
```



```

    }
    public void displayDetails() {
        System.out.println("Clothing -> " + name + " | Price: " + price + " | Size: " + size);
    }
}

class Grocery extends Product implements ProductInfo {
    String expiryDate;
    Grocery(String name, double price, String expiryDate) {
        super(name, price);
        this.expiryDate = expiryDate;
    }
    public void displayDetails() {
        System.out.println("Grocery -> " + name + " | Price: " + price + " | Expiry: " + expiryDate);
    }
}

public class JAVA_EXP_7 {
    public static void main(String[] args) {
        ProductInfo item1 = new Electronics("Smartphone", 25000, 1);
        ProductInfo item2 = new Clothing("T-Shirt", 599, "M");
        ProductInfo item3 = new Grocery("Rice Bag", 450, "Dec-2026");

        item1.displayDetails();
        item2.displayDetails();
        item3.displayDetails();
    }
}

```

```

PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS> java -cp .\bin\ 'JAVA_EXP_7'
Electronics -> Smartphone | Price: 25000.0 | Warranty: 1 yrs
Clothing -> T-Shirt | Price: 599.0 | Size: M
Grocery -> Rice Bag | Price: 450.0 | Expiry: Dec-2026
PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>

```